



Khalifa Bin Zayed School

Physics Department - Grade 12 Elite

Chapter 13: electromagnetic Induction - Transformer Simulation Lab

Subject: Physics
Grade: 12 Elite

Student Identification

Name:



Class:

Date:

Objective

To quantitatively verify the Transformer Law ($\frac{V_s}{V_p} = \frac{N_s}{N_p}$) and analyze the conservation of energy and power efficiency in step-up and step-down configurations.

Theoretical Framework

A transformer works on the principle of **Faraday's Law of Induction**. The changing magnetic flux created by the primary AC voltage induces an electromotive force (EMF) in the secondary coil.

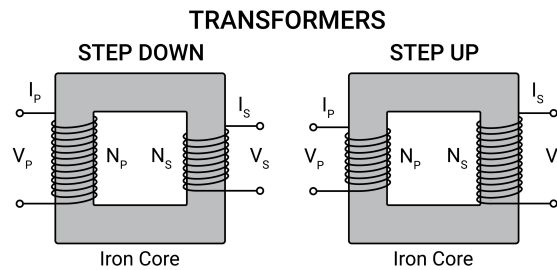


Figure 1: Transformer Schematic showing N_p , N_s , and the ferromagnetic core.

Experimental Data Collection

Trial	V_p (V)	N_p	N_s	Simulated V_s (V)	Theor. V_s ($V_p \cdot \frac{N_s}{N_p}$)
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Elite Analysis: Power, Energy & Efficiency

In an *ideal* transformer, $P_p = P_s$. In a real transformer, energy is lost primarily due to Joule heating and eddy currents.

Advanced Problem: Set $V_p = 220V$. Adjust turns to achieve $V_s = 110V$. If the measured $I_p = 0.5A$ and $I_s = 0.9A$:

- Calculate Input Power (P_{in}): W
- Calculate Output Power (P_{out}): W
- Determine Efficiency ($\eta = \frac{P_{out}}{P_{in}} \times 100\%$): %
- Energy lost as heat over 5 minutes ($E_{loss} = P_{loss} \times t$): J

Critical Thinking & Conclusion

Explain why the efficiency calculated above is less than 100% based on the physics of the magnetic core

shown in Figure 1.